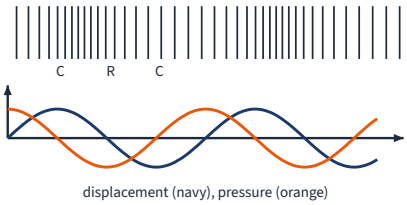
CHAPTER 21 SOUND WAVES

Key idea: sound is a longitudinal pressure wave; air columns and strings resonate at definite frequencies.

1 PROPAGATION OF SOUND

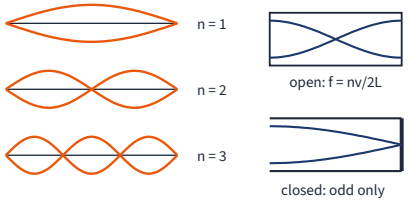


displacement (navy), pressure (orange)

$$v = \sqrt{\frac{\gamma RT}{M}} \qquad v = \sqrt{\frac{E}{\rho}}$$

- Longitudinal: compressions and rarefactions
- Needs a medium; faster in solids
- Gas: $v \propto \sqrt{T}$, independent of pressure

2 AIR COLUMNS



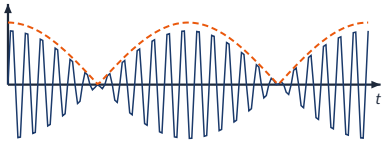
Pipe	Ends	Harmonics	f_1
Open	A–A	all	$v/2L$
Closed	N–A	odd only	$v/4L$

3 INTENSITY LEVEL

$$\beta = 10 \log_{10} \frac{I}{I_0}$$

- $I_0 = 1.0 \times 10^{-12}$ W m^{-2} (threshold of hearing)
- +10 dB = intensity ×10; +3 dB ≈ ×2
- N identical sources: $\beta + 10 \log_{10} N$
- Add intensities, **not** decibels

4 BEATS



loudness rises and falls $f_1 - f_2$ times per second

$$f_{\text{beat}} = |f_1 - f_2|$$

$$f = \frac{f_1 + f_2}{2}$$

Loudness rises and falls as the waves move in and out of phase.

5 DOPPLER EFFECT

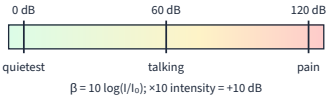
$$f' = f \frac{v \pm v_o}{v \mp v_s}$$

- Upper signs: **approaching** (frequency rises)
- Speeds relative to the air
- Moving reflector: apply twice (observer, then source)

6 STRINGS & END CORRECTION

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$$

- Node at each fixed end; all harmonics
- Closed pipe: $L + e = \frac{\lambda}{4}$
- Antinode lies slightly **beyond** the open end



7 UNIT CHECK

Quantity	Formula	Unit
Intensity	$P/4\pi r^2$	W m^{-2}
Intensity level	$10 \log(I/I_0)$	dB
Beat frequency	$\lvert f_1 - f_2 \rvert$	Hz
Mass per length	μ	kg m^{-1}

8 COMMON SECTION A TRAPS

- Closed pipe has **odd** harmonics only: 1st, 3rd, 5th…
- Adding dB values directly.
- Using °C in $v \propto \sqrt{T}$.
- Doppler: the wrong sign for approach.
- Displacement node = pressure **antinode**.

EXAM TIPS

- ✓Sketch the air column with N and A before using a formula.
- ✓Convert dB to intensity, add, then convert back.
- ✓Write f' formula with signs explained in words.
- ✓Use $v = 4f(L + e)$ for resonance-tube data.